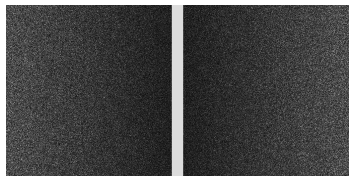


UNDERWATER SIDE-SCAN SONAR

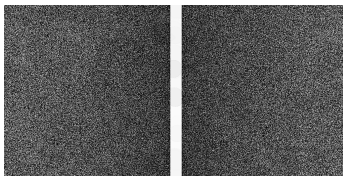
Debris & Anomaly Inspection Report

Survey Mission: MISSION-E2E | Record: E2E-TEST-SONAR-001
Generated: 2026-09-11 15:34:02 UTC

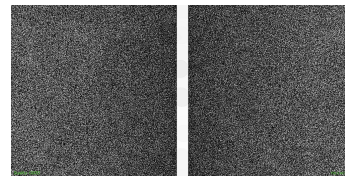
1. Acoustic Imagery Overview



RAW ACOUSTIC



ENHANCED & FILTERED



DETECTIONS & TARGETS

2. Signal Quality Analysis

Metric	Value
Quality Grade	GOOD
Composite Score	0.800
Sharpness (Laplacian var.)	32102.58
Contrast Deviation	51.25
Acoustic Noise Index	26.57
Mean Intensity	61.6
Missing Data Columns	0.0%
Observations	Elevated noise (26.6)

3. Processing Latency Breakdown

Pipeline Stage	Duration (ms)
Preprocessing & CLAHE	60.9
Object Detection	2940.3
Anomaly Detection	6.6
Acoustic Shadow Analysis	0.0
Total Latency	3039.6

4. Target Detections (0 target(s) identified)

No targets identified within threshold in this frame.

5. Acoustic Shadow Validation

No prominent acoustic shadow signatures identified for targets in this frame.

6. Seabed Anomaly Scoring

Attribute	Value
Anomaly Index	0.151
Anomalous Status	False
Evaluation Mode	REAL_AUTOENCODER
Contributing Factors	Reconstruction consistent with learned seabed distribution

7. Confidence Fusion Summary

8. Inspection System Architecture

Step	Pipeline Stage
1. Input Ingestion	Raw acoustic side-scan sonar image ingestion from survey datalog.
2. Quality Screening	Contrast, Laplacian variance, and transducer dropout assessment.
3. Adaptive Equalization	Contrast Limited Adaptive Histogram Equalization (CLAHE).
4. Acoustic Denoising	Bilateral / non-local means adaptive noise suppression.
5. Object Detection	YOLOv8 deep neural inference with fallback heuristic rules.
6. Geomorphology Filter	Differentiation between natural seabed features and debris.
7. Shadow Analysis	Validation of physical elevation through acoustic shadow dropout.
8. Anomaly Detection	Regional intensity and Shannon entropy deviation analysis.
9. Multi-Modal Fusion	Cross-signal weighted evidence synthesis.
10. Threat Severity	Operational risk categorization based on target dimensions and class.
11. Positional Tagging	Synchronization with vehicle navigation and GPS telemetry.
12. Report Dispatch	Tabular and spatial log generation for mission debrief.